Pablo Aganza

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Experience

Foresight Diagnostics Remote
Software Engineer, Data *July 2023 - January 2026*

- Reduced dataset preparation and validation from 7 days to under 1 hour by leading development of a scalable Python ETL pipeline on GCP with custom parsers and normalizers for varied input formats
- Accelerated new clinical data source integration from 2-3 days to under 2 hours of configuration by architecting a JSON config-driven layer that dynamically generated Pydantic validation schemas per data source
- Cut data issue investigation from 3-5 days to under 4 hours by implementing regulatory-compliant logging, monitoring, and audit trails with UI-integrated error surfacing for scientists to self-diagnose issues
- Saved 20+ software engineering hours per week by translating scientific-team requirements into self-service, FDA submission-ready export tooling that produced immutable, versioned outputs in GCS
- Scaled pipeline tool access from 1 operator to ~20 team members by deploying a Dockerized GCP Cloud Run service with CI/CD, replacing local execution with a reliable shared service

Software Engineer Intern *January 2023 - July 2023*

- Improved bioinformatics pipeline performance through threshold sensitivity analysis that supported a production threshold change

Genalyte San Diego, CA
Machine Learning Engineer Intern *May 2022 - July 2022*

- Automated blood diagnostic robot state classification by deploying a transfer-learning computer vision model to support R&D testing workflows and quality-control checks

Projects

Imminent Seizure Prediction (Master's Degree Capstone) Project Page

- Processed 572 GB of raw EEG seizure-monitoring data into a 75 GB model-ready dataset, creating labeled time-series sequences to predict seizures within the next 60 seconds
- Benchmarked supervised and unsupervised PyTorch models across patient-level splits, selecting an autoencoder for its classification performance and identifying individualized calibration as key to reliable seizure prediction

Flight Departure Delay Prediction Project Page

- Built a Spark pipeline integrating 660M+ flight and weather records, using forward-chaining cross-validation and scheduled-time joins to prevent temporal leakage for realistic two-hour-ahead delay forecasting
- Designed and implemented a snapshot-temporal graph neural network end-to-end (PyTorch Geometric, TransformerConv, Optuna) to model network-wide delay propagation, benchmarked against gradient-boosted trees

Education

University of California, Berkeley Berkeley, CA
Master of Information and Data Science; GPA: 3.92/4.00 *January 2024 - May 2026*

Carnegie Mellon University Pittsburgh, PA
B.S. Mathematical Sciences; GPA: 3.30/4.00 *August 2018 - May 2023*
Minor in Computer Science, concentration in Robotics

Technical Skills

Big Data / Distributed: Apache Spark, PySpark, Databricks, Parquet, GCP Pub/Sub, Hadoop, MapReduce
Data Platforms: PostgreSQL, Neo4j, Redis
Data Science Libraries: NumPy, SciPy, pandas, GeoPandas
Data Visualization / Monitoring: Matplotlib, Seaborn, Grafana
Languages: Python, SQL, R, Bash
Deep Learning: PyTorch, PyTorch Geometric, TensorFlow/Keras, Hugging Face Transformers
Classical ML: scikit-learn, XGBoost, MLlib
ML Tooling / MLOps: Optuna, FastAPI, Pydantic
Platforms / DevOps: GCP, AWS, Docker, Kubernetes, Git, GitHub Actions, CI/CD, pytest (unit/integration)